Q1.

The binary operation * is defined as

$$a * b = a + b + 4 \pmod 6$$

where $a, b \in \mathbb{Z}$

(a) Show that the set {0, 1, 2, 3, 4, 5} forms a group G under &42;.

(5)

(b) Find the proper subgroups of the group G in part **(a)**.

(2)

(c) Determine whether or not the group G in part **(a)** is isomorphic to the group $K = (\langle 3 \rangle, \times_{14})$

(3)

(Total 10 marks)